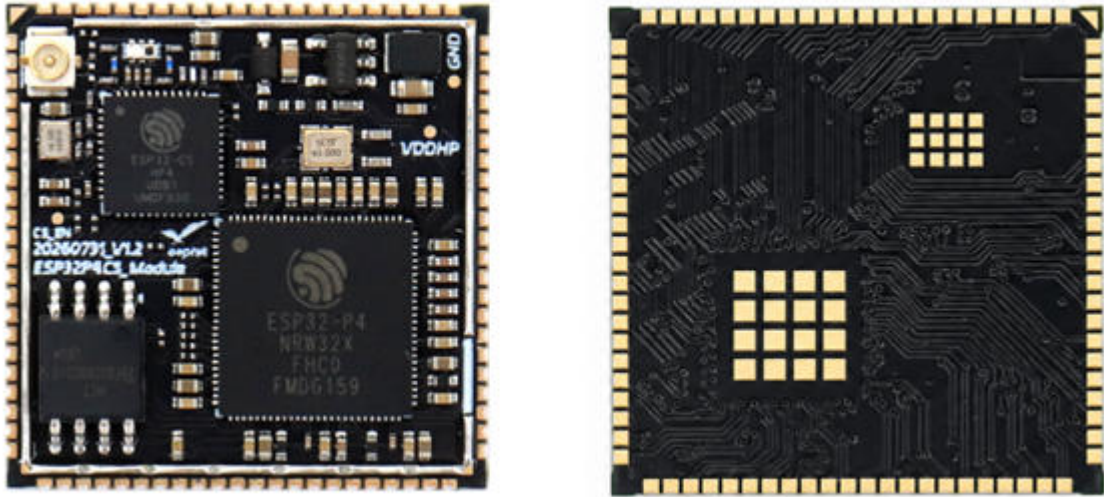




ESP32-P4C5-Core Development Board

Technical Specifications

Revision History

Version	Date	Content created/amended	Author/Reviser	Reviewer

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1. Overview

1.1. Product Overview

The ESP32-P4C5-Core is a compact, pin-pitch core board with integrated NOR FLASH, designed based on the Espressif ESP32-P4 chip. The core processor chip, the ESP32-P4, features 32MB of on-chip PSRAM and comprises two high-performance (HP) cores and one low-power (LP) core. The HP cores utilise a RISC-V dual-core processor with a clock speed of up to 400MHz, incorporating a JPEG codec, a pixel processing accelerator,

an H.264 video encoder and a MIPI interface; it offers powerful image and voice processing capabilities.

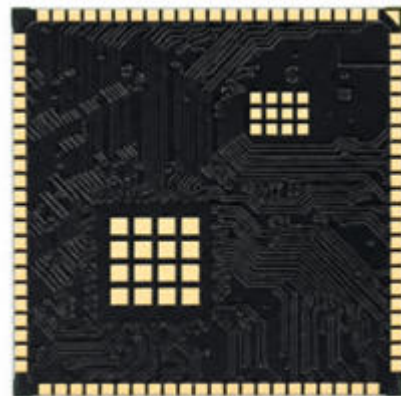
1.2. Features

- Dual-core CPU with a high clock speed of 400 MHz.
- Built-in 16 MB Flash and 32 MB PSRAM.
- Full pin-out of the ESP32-P4 chip.
- The core board is compact, facilitating hardware design
- Comprehensive development resources

1.3. Product images



Front view



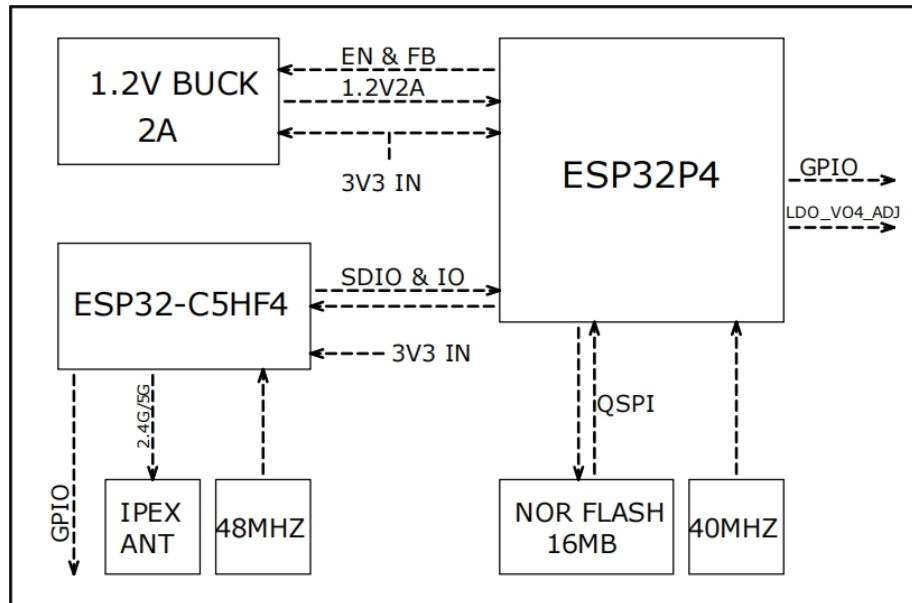
Rear

1.4. Application scenarios

- Smart Home
- Industrial Automation
- Consumer Electronics
- HMI (Human-Machine Interface)
- Robotics
- Camera video streaming
- USB devices

2. Product Specifications

2.1. Function Block Diagram



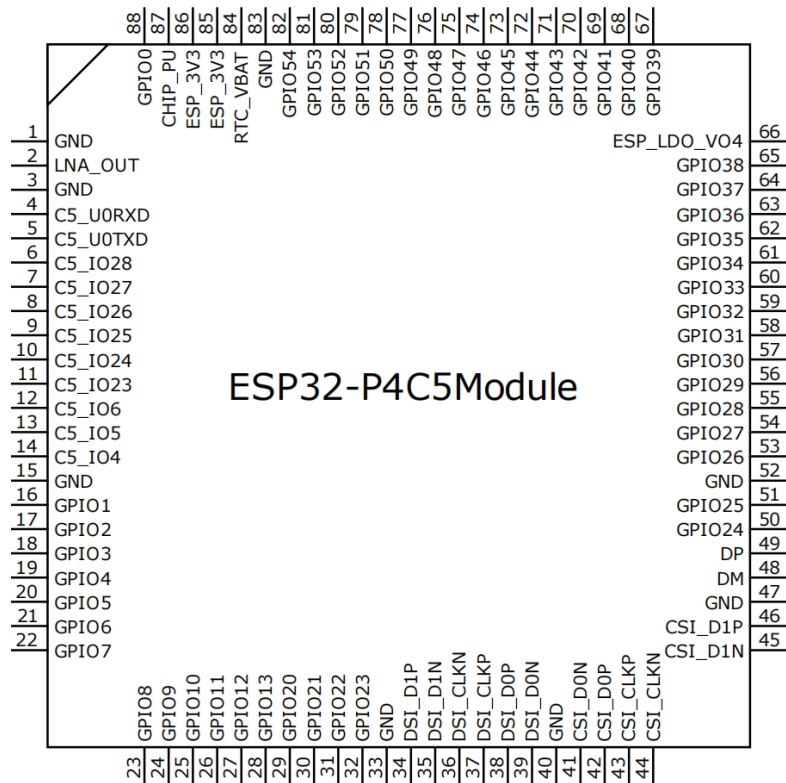
2.2 Hardware Specifications

Main controller	CPU	ESP32-P4
	Core	RISC-V 32-bit dual-core processor
	Clock speed	400 MHz (HP system)
		40 MHz (LP system)
Memory	ROM	128 KB HP ROM
		16 KB LP ROM
	SRAM	768 KB HP L2MEM
		32 KB LP SRAM
	Flash	16 MB
Peripheral interfaces	GPIO	55
	SPI	2
	LP SPI	1
	UART	5

	LP UART	1
	I3C	1
	I2C	2
	LP I2C	1
	I2S	3
	LP I2S	1
	USB JTAG	1
	SDIO	1
	LED PWM	1
	MCPWM	2
	TWAI Controller (ISO 11898-1 compliant)	3
	High-speed USB 2.0 OTG	1
	Full-Speed USB 2.0 OTG	1
	100 Mbps Ethernet MAC	1
	MIPI CST-2	1
	MIPI DSI	1
	Parallel I/O (PARLIO) Controller	1
	12-bit multi-channel analogue-to-digital converter	2
	Temperature sensor	1
	Touch sensor	1
	Analogue voltage comparator	1
	Undervoltage monitoring	1
Image and Audio Processing Interface	JPEG Encoder/Decoder	1
	Pixel Processing Accelerator (PPA)	1
	Image Signal Processor (ISP)	1
	H.264 Video Encoder	1

3. Pin Definitions

3.1. Pin Layout



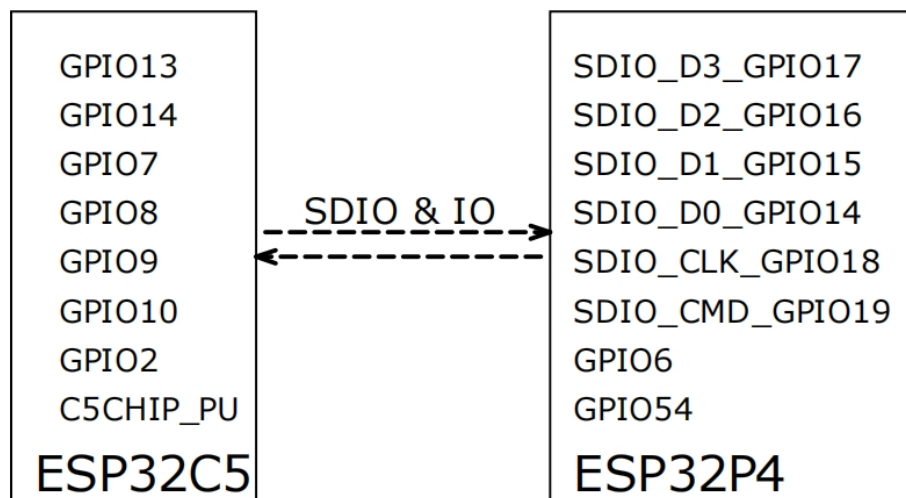
3.2. Pin Description

Pin	Name	Description
1	GND	Power ground
2	LAN_OUT	ESP32-C5 ANT antenna interface (on-board IPEX-1 interface)
3	GND	Power ground
4	C5_U0RXD	ESP32-C5 U0RXD (ESP32-C5 programming pin)
5	C5_U0TXD	ESP32-C5 U0TXD (ESP32-C5 programming pin)
6	C5_IO28	GPIO28 on the ESP32-C5 (ESP32-C5 strapping pin)
7	C5_IO27	GPIO27 on the ESP32-C5 (ESP32-C5 strapping pin)
8	C5_IO26	GPIO26 on the ESP32-C5 (ESP32-C5 strapping pin)

9	C5_IO25	GPIO25 on the ESP32-C5
10	C5_IO24	GPIO24 on the ESP32-C5
11	C5_IO23	GPIO23 on the ESP32-C5
12	C5_IO6	GPIO6 on the ESP32-C5
13	C5_IO5	GPIO5 on the ESP32-C5
14	C5_IO4	GPIO4 on the ESP32-C5
15	GND	Ground
16	GPIO1	GPIO1, LP_GPIO1, XTAL_32K_P
17	GPIO2	GPIO2, MTCK, LP_GPIO2, TOUCH_CHANNEL0
18	GPIO3	GPIO3, MTDI, LP_GPIO3, TOUCH_CHANNEL1
19	GPIO4	GPIO4, MTMS, LP_GPIO4, TOUCH_CHANNEL2
20	GPIO5	GPIO5, MTDO, LP_GPIO5, TOUCH_CHANNEL3
21	GPIO6	GPIO6, SPI2_HOLD_PAD, LP_GPIO6, TOUCH_CHANNEL4
22	GPIO7	GPIO7, SPI2_CS_PAD, LP_GPIO7, TOUCH_CHANNEL5
23	GPIO8	GPIO8, UART0_RTS_PAD, SPI2_D_PAD, LP_GPIO8, TOUCH_CHANNEL6
24	GPIO9	GPIO9, UART0_CTS_PAD, SPI2_CK_PAD, LP_GPIO9, TOUCH_CHANNEL7
25	GPIO10	GPIO10, UART1_TXD_PAD, SPI2_Q_PAD, LP_GPIO10, TOUCH_CHANNEL8
26	GPIO11	GPIO11, UART1_RXD_PAD, SPI2_WP_PAD, LP_GPIO11, TOUCH_CHANNEL9
27	GPIO12	GPIO12, LP_GPIO12, TOUCH_CHANNEL10
28	GPIO13	GPIO13, LP_GPIO13, TOUCH_CHANNEL11
29	GPIO20	GPIO20, ADC1_CHANNEL4
30	GPIO21	GPIO21, ADC1_CHANNEL5
31	GPIO22	GPIO22, ADC1_CHANNEL6
32	GPIO23	GPIO23, ADC1_CHANNEL7, REF_50M_CLK_PAD
33	GND	Power ground
34	DSI_D1P	MIPI DSI PHY DATAP1
35	DSI_D1N	MIPI DSI PHY DATAN1
36	DSI_CLKN	MIPI DSI PHY CLKN
37	DSI_CLKP	MIPI DSI PHY CLKP
38	DSI_D0P	MIPI DSI PHY DATAP0
39	DSI_D0N	MIPI DSI PHY DATAN0
40	GND	Power Ground
41	CSI_D0N	MIPI CSI PHY DATA N

42	CSI_D0P	MIPI CSI PHY DATA0
43	CSI_CLKP	MIPI CSI PHY CLKP
44	CSI_CLKN	MIPI CSI PHY CLKN
45	CSI_D1N	MIPI CSI PHY DATAN1
46	CSI_D1P	MIPI CSI PHY DATAP1
47	GND	Power Ground
48	DM	USB 2.0 OTG PHY DM
49	DP	USB 2.0 OTG PHY DP
50	GPIO24	GPIO24, USB1P1_N0
51	GPIO25	GPIO25, USB1P1_P0
52	GND	Power ground
53	GPIO26	GPIO26, USB1P1_N1
54	GPIO27	GPIO27, USB1P1_P1
55	GPIO28	GPIO28, SPI2_CS_PAD, GMAC_PHY_RXDV_PAD
56	GPIO29	GPIO29, SPI2_D_PAD, GMAC_PHY_RXD0_PAD
57	GPIO30	GPIO30, SPI2_CK_PAD, GMAC_PHY_RXD1_PAD
58	GPIO31	GPIO31, SPI2_Q_PAD, GMAC_PHY_RXER_PAD
59	GPIO32	GPIO32, SPI2_HOLD_PAD, GMAC_RMII_CLK_PAD
60	GPIO33	GPIO33, SPI2_WP_PAD, GMAC_PHY_TXEN_PAD
61	GPIO34	GPIO34, SPI2_IO4_PAD, GMAC_PHY_TXD0_PAD
62	GPIO35	GPIO35, SPI2_IO5_PAD, GMAC_PHY_TXD1_PAD (no pull-up inside the module)
63	GPIO36	GPIO36, SPI2_IO6_PAD, GMAC_PHY_TXER_PAD (no pull-up inside the module)
64	GPIO37	GPIO37, UART0_TXD_PAD, SPI2_IO7_PAD (ESP32-P4 programming pin)
65	GPIO38	GPIO38, UART0_RXD_PAD, SPI2_DQS_PAD (ESP32-P4 programming pin)
66	ESP_LDO_VO4	Output power supply (output voltage range 0.5–2.7 V or 3.3 V, maximum output current 0.2 A)
67	GPIO39	GPIO39, SD1_CDATA0_PAD, REF_50M_CLK_PAD
68	GPIO40	GPIO40, SD1_CDATA1_PAD, GMAC_PHY_TXEN_PAD
69	GPIO41	GPIO41, SD1_CDATA2_PAD, GMAC_PHY_TXD0_PAD
70	GPIO42	GPIO42, SD1_CDATA3_PAD, GMAC_PHY_TXD1_PAD
71	GPIO43	GPIO43, SD1_CCLK_PAD, GMAC_PHY_TXER_PAD
72	GPIO44	GPIO44, SD1_CCMD_PAD, GMAC_RMII_CLK_PAD
73	GPIO45	GPIO45, SD1_CDATA4_PAD, GMAC_PHY_RXDV_PAD
74	GPIO46	GPIO46, SD1_CDATA5_PAD, GMAC_PHY_RXD0_PAD
75	GPIO47	GPIO47, SD1_CDATA6_PAD, GMAC_PHY_RXD1_PAD

76	GPIO48	GPIO48, SD1_CDATA7_PAD, GMAC_PHY_RXER_PAD
77	GPIO49	GPIO49, GMAC_PHY_TXEN_PAD, ADC2_CHANNEL0
78	GPIO50	GPIO50, GMAC_RMII_CLK_PAD, ADC2_CHANNEL1
79	GPIO51	GPIO51, GMAC_PHY_RXDV_PAD, ADC2_CHANNEL2, ANA_COMP0
80	GPIO52	GPIO52, GMAC_PHY_RXD0_PAD, ADC2_CHANNEL3, ANA_COMP0
81	GPIO53	GPIO53, GMAC_PHY_RXD1_PAD, ADC2_CHANNEL4, ANA_COMP1
82	GPIO54	GPIO54, GMAC_PHY_RXER_PAD, ADC2_CHANNEL5, ANA_COMP1
83	GND	Power ground
84	RTC_VBAT	The backup power supply pin must not be left floating; the voltage range is 2.3V to 3.6V (can be connected to a 3.3V supply)
85	ESP_3V3	Power supply (3.3V input from the core board)
86	ESP_3V3	Power supply (3.3V input from core board)
87	CHIP_PU	Enable ESP32-P4 chip (no internal RC reset; external reset required)
88	GPIO0	GPIO0, LP_GPIO0, XTAL_32K_N



Pin mapping between ESP32C5HF4 and ESP32P4

3.3. Boot options configuration

3.3.1. Strapping

Upon power-up or hardware reset, the following boot parameters can be configured via the Strapping pins and eFuse bits without the involvement of the microprocessor:

- Chip boot mode

- Strapping Pins: GPIO35, GPIO36, GPIO37, GPIO38
- ROM Logging
- Strapping pins: GPIO36
- eFuse bit: EFUSE UART PRINT CONTROL
- JTAG Signal Source
- Strapping pins: GPIO34
- eFuse bits: EFUSE_DIS_PAD_JTAG, EFUSE_DIS_USB_JTAG and EFUSE_JTAG_SEL_ENABLE The default values for the above eFuse

bits are all set to 0 by default, meaning they have not been programmed.

eFuses can only be programmed once; once set to 1, they cannot be restored to 0.

If the aforementioned strapping pins are not connected to any circuit or the connected circuit is in a high-impedance state, their default value (i.e. logic level) depends on the state of the internal weak pull-up/pull-down resistor at reset.

Default configuration of strapping pins

Strapping Pins	Default Configuration	Value
GPIO34	Floating	-
GPIO35	Weak pull-up	1
GPIO36	Floating	-
GPIO37	Floating	-
GPIO38	Floating	-

To change the value of the strapping pins, external pull-down or pull-up resistors can be connected. If the ESP32-P4 is used as a slave device to a host MCU, the level of the strapping pins can also be controlled via the host MCU.

All strapping pins have latches. Upon system reset, the latch samples and stores the value of the corresponding strapping pin, which is retained until the chip is powered down or switched off. The state of the latch cannot be altered by any other means. Consequently, the value of the strapping pin remains readable whilst the chip is active, and the strapping pin functions as a standard I/O pin after the chip is reset.

3.3.2. Chip Start-up Mode Control

After reset is released, GPIO35 to GPIO38 collectively determine the start-up mode. See the table below for details.

Start-up Mode	GPIO35	GPIO36	GPIO37	GPIO38
SPI Boot*	1*	Any value	Any value	Any value
Joint Download Boot	0	1	Any value	Any value

Indicates the default value and default configuration.

The following download methods are supported in Joint

Download Boot mode: USB Download Boot:

- USB-Serial-JTAG Download Boot
- USB 2.0 OTG Download Boot

- UART Download Boot
- SPI Slave Download Boot

3.3.3. ROM- Log Printing Control

During system boot, the ROM code log can be printed to:

- (default) UART0 and USB serial/JTAG controller
- USB serial port/JTAG controller
- UART0

EFUSE_UART_PRINT_CONTROL and GPIO36 control UART0 ROM log printing; see the table below

UART ROM log output	EFUSE_UART_PRINT_CONTROL	GPIO37
Enable	0*	Ignore
	1	0
	2	1
Close	1	1
	2	0
	3	Ignore

* Indicates default values and default configuration.

EFUSE_DIS_USB_SERIAL_JTAG_ROM_PRINT controls USB serial/JTAG controller ROM logging; see the table below for details.

USB Serial/JTAG ROM Logging Control	EFUSE_DIS_USB_SERIAL_JTAG_ROM_PRINT
Enable*	0*
Disabled	1

*Indicates default value and default configuration

4. Electrical characteristics

4.1. Absolute Maximum Limits

Exceeding the absolute maximum ratings may result in permanent damage to the device. These are merely highlighted ratings and do not imply functional operation of the device under these or other conditions beyond the specifications. Prolonged exposure to absolute maximum conditions may affect the reliability of the ESP32-P4C5-Core.

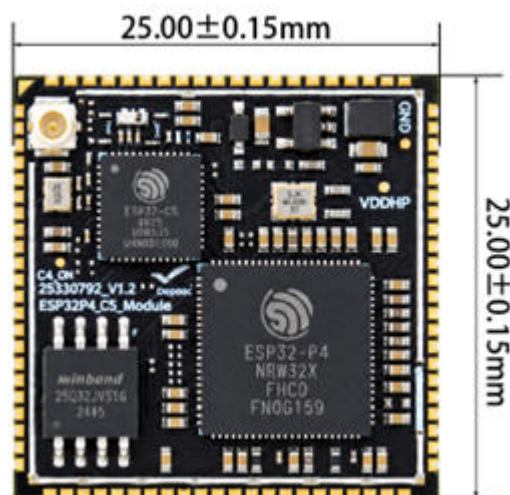
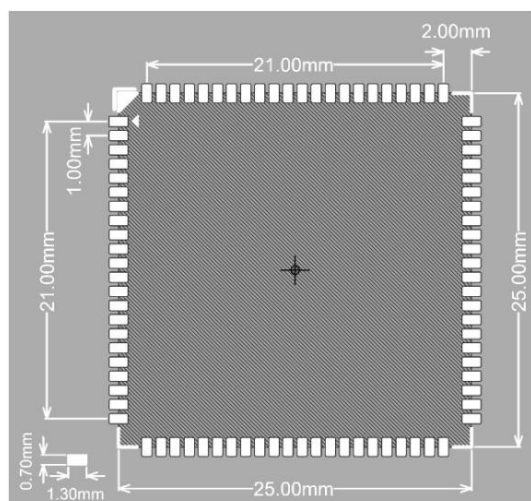
4.2. Power Consumption Characteristics

Symbol	Parameter	Minimum	Typical	Maximum	Unit
VCC	Power supply pin voltage	3.0	3.3	3.6	V
IVCC	External power supply current	-	1	-	A
TA	Operating temperature	-40	-	85	°C

5. Schematic

Not available

6. Product dimensions



7. Storage Conditions

Conditions	Specifications
Storage conditions	In a sealed MBB, in a non-condensing atmospheric environment at <40°C/90% RH
Operating conditions	At 25±5 °C and 60% RH for 168 hours
Moisture sensitivity	Grade 3

8. Reflow soldering profile

